

Erdős Problem #820

EP-820

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $H(n)$ be the smallest integer l such that there exist $k < l$ with $(k^n - 1, l^n - 1) = 1$.

Is it true that $H(n) = 3$ infinitely often? (That is, $(2^n - 1, 3^n - 1) = 1$ infinitely often?)

Estimate $H(n)$. Is it true that there exists some constant $c > 0$ such that, for all $\epsilon > 0$, $H(n) > \exp(n^{(c-\epsilon)/\log \log n})$ for infinitely many n and $H(n) < \exp(n^{(c+\epsilon)/\log \log n})$ for all large enough n ?

Does a similar upper bound hold for the smallest k such that $(k^n - 1, 2^n - 1) = 1$?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>